

SuperRS: Multi Scenario Reciprocal-Aware Dual MoE for Unified Recommendation-Search Ranking

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Abstract

In e-commerce, search and recommendation rankings require a deep understanding of user behaviors and personalized scoring of products. While existing systems maintain separate pipelines for search and recommendation, these two scenarios share aligned objectives and exhibit consistent data patterns during ranking. To address this, we propose a joint modeling approach for search-recommendation ranking that enables information gain exchange between the two scenarios, thus facilitating enhanced modeling of users' cross-scenario behaviors. Our proposed SuperRS framework employs a Dual-layer Multi-MoE (DualMoE) architecture to tackle scenario-specific disparities and achieve multi-interest fusion perception. A key aspect is the Search-Recommendation Sequence Fusion Unit, which integrates user interaction sequences from both scenarios. Additionally, we introduce a unified Representation Extraction method utilizing Reciprocal Scenario Interest Attention (RSIA) for feature alignment. Dynamic Feature Integration (DFI) employs a dual gating mechanism for controlled information fusion while preserving scenario identities, combined with multi-objective optimization. On the 1688 App, our framework demonstrates superior performance to baseline models across both offline evaluation metrics and online business indicators.

CCS Concepts

• **Information systems** → **Retrieval models and ranking.**

Keywords

Recommendation-Search Ranking, Dual MoE, Scenario Disparities, Behavior Fusion, Attention Alignment

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1 Introduction

Recommendation and Search (RS) functions have emerged as crucial services for online applications, prompting extensive research aimed at improving their effectiveness [6, 8, 11, 16, 17, 23]. The integration of RS systems has become pivotal for information services, leveraging their symbiotic capture of complementary user intent [20, 22]. While recommendation systems infer latent preferences via historical interactions, search engines address explicit, transient needs. While there are many valid approaches to multi-scenario multi-targeting [4, 7, 15, 19], RS unification still mitigates persistent challenges: recommendation cold-start from data sparsity and search degradation due to query ambiguity [12, 13]. Crucially, search behaviors encode real-time intent, whereas recommendations reflect longitudinal interests, forming a causal-complementary dyad [18, 21]. Key to this synergy is disentangling user representations across behavioral domains. Research spans hierarchical embeddings modeling intent transitions [1] to GNNs unifying interactions in heterogeneous graphs [5, 24]. Notable advances include cross-domain decoupling to eliminate search bias [13], counterfactual causal frameworks using search as instrumental variables [12], and unified sequential models integrating temporal attention [20]. Also, pareto-optimization frameworks further balance conflicting objectives [21].

Conventional multi-scenario models face two critical limitations in recommendation and search (RS) scenarios. First, they fail to effectively capture scenario-specific differences and behavioral nuances, leading to a shallow understanding of user interactions and reduced personalization. Second, these models underutilize contextual product features and implicit information from search and recommendation sequences, missing opportunities to improve prediction accuracy and relevance. These issues result in oversimplified user behavior interpretations, neglecting cross-platform complexities and compromising RS integration performance.

To address these challenges, we propose a hierarchical framework combining dual-layer mixture-of-experts (DualMoE) with bidirectional cross-scenario attention. The design features two interdependent modules: context-aware hierarchical experts for progressive feature abstraction and Reciprocal Scenario Interest Attention (RSIA) for synergistic interaction modeling. The first MoE layer employs scenario-specific expert networks to preserve domain characteristics while establishing cross-scenario connections via a compatibility-driven gating mechanism. The second MoE layer processes enhanced behavioral sequences from RSIA, integrating sequence semantics, contextual encodings, and item-specific relevance through target-aware expert selection. The key contributions of this paper are as follows:

- **Dual-Layer Mixture-of-Experts (DualMoE):** Captures scenario-specific characteristics and complex interactions between search and recommendation tasks at different abstraction levels.
- **Reciprocal Scenario Interest Attention (RSIA):** Facilitates bidirectional information flow between search and recommendation scenarios, enhancing user intent modeling by integrating cross-scenario interactions, target item relevance, and historical behavior context.
- **Dynamic Feature Integration (DFI) and Multi-Objective Optimization:** Employs dual gating for controlled information fusion while preserving scenario identities, combined with a multi-objective optimization strategy to balance loss components and enhance model performance.

2 Problem Formulation

The joint modeling of search and recommendation scenarios constitutes a multi-faceted learning task defined over a unified interaction space $\mathcal{D} = \{(Q_s, C_r, \mathbf{t}, y)\}$, where $Q_s = \{\mathbf{q}_i\}_{i=1}^{L_s}$ represents search session features with $\mathbf{q}_i \in \mathbb{R}^{d_q}$ denoting d_q -dimensional query embeddings, $C_r = \{\mathbf{c}_j\}_{j=1}^{L_r}$ encapsulates recommendation context features through d_c -dimensional item context vectors $\mathbf{c}_j \in \mathbb{R}^{d_c}$, $\mathbf{t} \in \mathbb{R}^{d_t}$ specifies the target item embedding requiring engagement prediction, and $y \in \{0, 1\}$ indicates observed user interaction labels. The learning objective centers on deriving a unified predictive function $f_\theta : Q_s \times C_r \times \mathbb{R}^{d_t} \rightarrow [0, 1]$.

3 Method

Our framework introduces a dual-layer mixture-of-experts (MoE) architecture coupled with a novel cross-scenario attention mechanism, designed to jointly model search and recommendation tasks while preserving scenario-specific characteristics.

3.1 Hierarchical MoE Architecture

The dual-layer MoE structure processes features at different abstraction levels through specialized expert networks. The first layer focuses on raw scenario-specific feature encoding, while the second layer handles cross-scenario sequence fusion.

3.1.1 Context-Aware Feature Encoding (Layer-1). The primary MoE layer processes raw input features through scenario-specific experts.

Given a search query embedding $\mathbf{q}_{\text{search}} \in \mathbb{R}^{d_q}$ and recommendation context vector $\mathbf{c}_{\text{rec}} \in \mathbb{R}^{d_c}$, the context gating mechanism computes expert weights via learnable projection vectors $\mathbf{w}_m \in \mathbb{R}^{d_q+d_c}$:

$$G_m^{\text{ctx}} = \frac{\exp(\mathbf{w}_m^\top [\mathbf{q}_{\text{search}}; \mathbf{c}_{\text{rec}}])}{\sum_{m'=1}^M \exp(\mathbf{w}_{m'}^\top [\mathbf{q}_{\text{search}}; \mathbf{c}_{\text{rec}}])} \quad (1)$$

The aggregated context representation $\mathbf{h}^{\text{ctx}} \in \mathbb{R}^{d_h}$ combines outputs from M parallel experts $\{E_m^{\text{ctx}}\}$, each processing the concatenated scenario features:

$$\mathbf{h}^{\text{ctx}} = \sum_{m=1}^M G_m^{\text{ctx}} \cdot E_m^{\text{ctx}}([\mathbf{q}_{\text{search}}; \mathbf{c}_{\text{rec}}]) \quad (2)$$

This layer preserves scenario identity through dedicated experts while establishing initial cross-scenario feature interactions. The gating mechanism automatically emphasizes relevant experts based on input feature compatibility.

3.1.2 Cross-Scenario Sequence Fusion (Layer-2). Building upon the context encoding, the secondary MoE layer processes RSIA-enhanced sequence features $\mathbf{H}^{\text{RSIA}} \in \mathbb{R}^{L \times d_h}$ through adaptive expert selection. The gating network integrates three information sources: processed sequence features, context encoding $\mathbf{h}^{\text{ctx}}$, and target item embedding $\mathbf{t}_{\text{item}} \in \mathbb{R}^{d_t}$:

$$G_n^{\text{seq}} = \sigma(\mathbf{W}_g[\mathbf{H}^{\text{RSIA}}; \mathbf{h}^{\text{ctx}}; \mathbf{t}_{\text{item}}]) \quad (3)$$

where $\mathbf{W}_g \in \mathbb{R}^{N \times (d_h+d_c+d_t)}$ projects the concatenated features to N expert gates. The final sequence encoding $\mathbf{h}^{\text{seq}} \in \mathbb{R}^{d_h}$ aggregates outputs from sequence experts $\{E_n^{\text{seq}}\}$:

$$\mathbf{h}^{\text{seq}} = \sum_{n=1}^N G_n^{\text{seq}} \cdot E_n^{\text{seq}}(\mathbf{H}^{\text{RSIA}}) \quad (4)$$

This hierarchical design enables gradual feature abstraction, where Layer-1 captures scenario-specific patterns and Layer-2 models complex cross-scenario interactions.

The final prediction integrates multi-level representations through a Dual MoE:

$$\hat{y} = f_{\text{MoE}}([\mathbf{h}^{\text{com}}; \mathbf{h}^{\text{ctx}}; \mathbf{h}^{\text{seq}}; \mathbf{H}^{\text{RSIA}}]) \quad (5)$$

The architecture maintains dimension consistency across components: common features encoding $\mathbf{h}^{\text{com}}$, context encoding $\mathbf{h}^{\text{ctx}}$ and sequence encoding $\mathbf{h}^{\text{seq}}$, while RSIA outputs preserve original scenario feature dimensions for residual connection integrity.

3.2 Reciprocal Scenario Interest Attention

The Reciprocal Scenario Interest Attention (RSIA) mechanism facilitates bidirectional information flow between search and recommendation scenarios through dual gating.

3.2.1 Target-Aware Composite Attention. The proposed composite attention mechanism integrates three fundamental components to model scenario-specific user intent: (1) cross-scenario interaction patterns, (2) current target item relevance, and (3) historical behavior context. The enhanced attention computation operates through

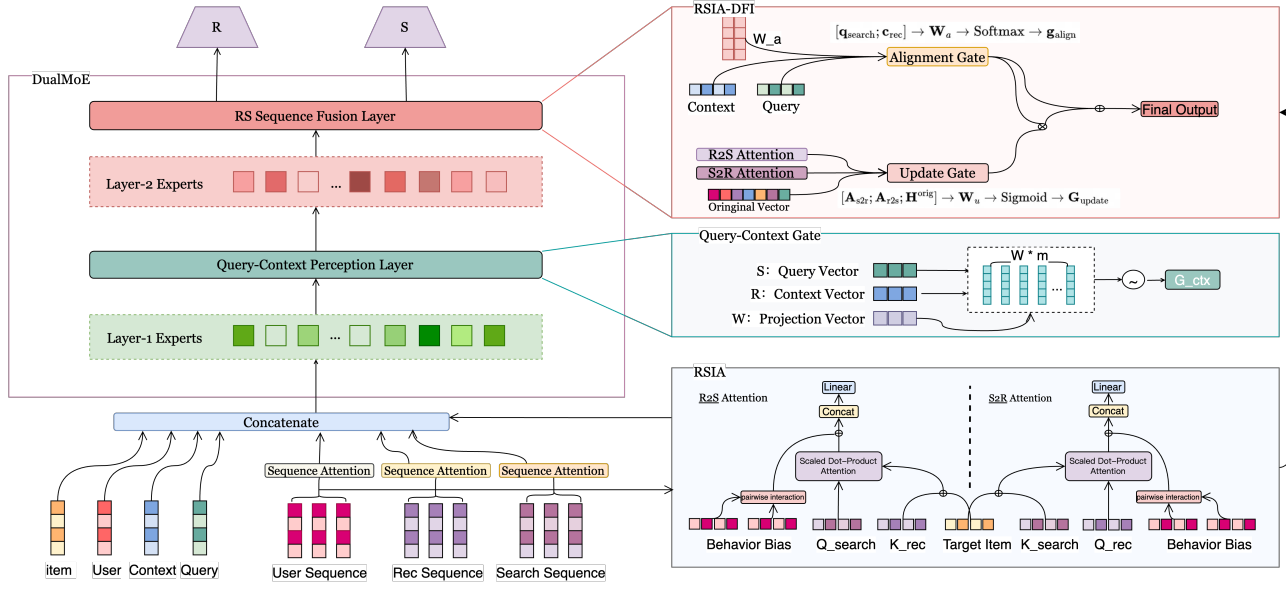


Figure 1: The overall architecture of the proposed model. The figure illustrates the key components, including the Reciprocal Scenario Interest Attention (RSIA) mechanism and the Dual MoE structure.

dimension-aware feature transformations:

$$A_{s2r} = \text{Softmax} \left(\frac{Q_{\text{search}} [K_{\text{rec}} \oplus f_{\text{proj}}(t_{\text{item}})]^T}{\sqrt{d}} + B_{\text{hist}} \right) \quad (6)$$

$$A_{r2s} = \text{Softmax} \left(\frac{Q_{\text{rec}} [K_{\text{search}} \oplus f_{\text{proj}}(t_{\text{item}})]^T}{\sqrt{d}} + B_{\text{hist}} \right) \quad (7)$$

where $t_{\text{item}} \in \mathbb{R}^{d_t}$ denotes the embedding of the target item being scored, $f_{\text{proj}} : \mathbb{R}^{d_t} \rightarrow \mathbb{R}^d$ represents a learnable linear projection aligning target item features with the attention space, and $B_{\text{hist}} \in \mathbb{R}^{L \times L}$ encodes historical behavior biases through pairwise interaction modeling:

$$B_{\text{hist}}[i, j] = \text{Sigmoid}(h_i^T W_b h_j) \quad (8)$$

Here $h_i \in \mathbb{R}^{d_h}$ corresponds to the i -th historical behavior embedding from the user's interaction sequence. The $\oplus$ operator indicates feature-wise addition with dimension broadcasting. where $t_{\text{item}} \in \mathbb{R}^{d_t}$ denotes the embedding of the target item being scored, $f_{\text{proj}} : \mathbb{R}^{d_t} \rightarrow \mathbb{R}^d$ represents a learnable linear projection aligning target item features with the attention space, and $B_{\text{hist}} \in \mathbb{R}^{L \times L}$ encodes historical behavior biases through pairwise interaction modeling. The learnable matrix $W_b \in \mathbb{R}^{d_h \times d_h}$ establishes latent interaction patterns between historical behaviors, the bilinear form $h_i^T W_b h_j$ measures compatibility between behavior i and j in projected space.

3.2.2 Dynamic Feature Integration (DFI). A two-stage gating mechanism controls information fusion. The update gate $G_{\text{update}} \in [0, 1]^{L \times d}$, parameterized by $W_u \in \mathbb{R}^{3d}$, determines cross-scenario feature injection ratios:

$$G_{\text{update}} = \text{Sigmoid} \left(W_u [A_{s2r}; A_{r2s}; H^{\text{orig}}] \right) \quad (9)$$

where $H^{\text{orig}} \in \mathbb{R}^{L \times d}$ denotes the original scenario-specific feature representations before cross-scenario fusion, serving as the baseline input to prevent over-distortion by attention mechanisms. The alignment gate $g_{\text{align}} \in \mathbb{R}^2$ preserves scenario identity through direct computation from raw features:

$$g_{\text{align}} = \text{Softmax} (W_a [q_{\text{search}}; c_{\text{rec}}]) \quad (10)$$

In recommendation scenario, final RSIA outputs combine updated and original features through residual connections:

$$H^{\text{RSIA}} = g_{\text{align}}[0] \cdot (G_{\text{update}} \odot W_u A_{s2r} V_{\text{rec}}) + g_{\text{align}}[1] \cdot [q_{\text{search}}; c_{\text{rec}}] \quad (11)$$

This dual gating ensures controlled information fusion while maintaining scenario identity, crucial for joint modeling of search and recommendation tasks.

3.3 Multi-Objective Optimization

The training objective combines three complementary losses through learnable weights λ_i :

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{task}} + \lambda_2 \mathcal{L}_{\text{align}} + \lambda_3 \mathcal{L}_{\text{gate}} \quad (12)$$

The task loss $\mathcal{L}_{\text{task}}$ employs label smoothing to prevent overconfidence in click-through rate prediction:

$$\mathcal{L}_{\text{task}} = - \sum_{i=1}^N \left[(1 - \alpha) y_i + \frac{\alpha}{K} \right] \log \sigma(\hat{y}_i) \quad (13)$$

where α controls smoothing intensity and K denotes binary classes. The alignment loss $\mathcal{L}_{\text{align}}$ enforces cross-scenario consistency through feature distance minimization and cosine similarity maximization:

$$\mathcal{L}_{\text{align}} = \|H^{\text{search}} - H^{\text{rec}}\|_F^2 + \left(1 - \frac{q_{\text{search}}^T c_{\text{rec}}}{\|q_{\text{search}}\| \|c_{\text{rec}}\|} \right) \quad (14)$$

Gate regularization $\mathcal{L}_{\text{gate}}$ controls expert activation sparsity to prevent over-specialization:

$$\mathcal{L}_{\text{gate}} = \frac{1}{M} \sum_{m=1}^M (G_m^{\text{ctx}} - \tau)^2 + \frac{1}{N} \sum_{n=1}^N (G_n^{\text{seq}} - \nu)^2 \quad (15)$$

τ and ν defining target activation levels. The adaptive weighting mechanism automatically balances loss components during training.

4 Experiment

4.1 Experimental Setup

Datasets and Evaluation Metrics. We evaluate our framework on two large-scale industrial datasets from Alibaba’s e-commerce platform: (1) **Ali_Search** containing 158 million search sessions, and (2) **Ali_Rec** comprising 132 million recommendation sessions. Following industrial practice, we adopt two-level evaluation: **Offline Metrics:** AUC and Loss evaluated separately on the Ali_Search and Ali_Rec datasets. **Online Metrics:** Click-Through Rate (CTR%), Conversion Rate (CVR%), Gross Merchandise Value (GMV) and User Average GMV (U-GMV) uplift.

Baselines. We compare against four state-of-the-art approaches:

- **ShareBottom** [2]: Classic multi-task framework with shared base layers and scenario-specific towers;
- **MMoE** [9]: Multi-gate MoE with scenario-adaptive gating for soft parameter sharing;
- **PLE** [14]: Layered expert routing separating shared/specific features to reduce conflicts;
- **STAR** [10]: Sparse attention with scenario-specific bias for memory-efficient adaptation;
- **PEPNet** [3]: Dual-gated scenario alignment with expandable module design.

This selection covers spectrum from classical multi-task models to latest multi-scenario approaches, enabling thorough analysis of both architectural evolution and performance boundaries in cross-scenario ranking tasks.

Implementation Details. Our DualMoE architecture employs two-layer expert networks with 8 experts per layer, each implemented as a three-layer MLP (512-256-128 dimensions). The RSIA module utilizes 4 attention heads with 64-dimensional projections. Training leverages the AdamW optimizer ($\text{lr}=1\text{e-}3$, $\beta_1=0.9$, $\beta_2=0.95$) for parameter updates.

4.2 Results and Discussion

4.2.1 Offline Evaluation. As quantitatively demonstrated in Table 1, SuperRS establishes new state-of-the-art performance across both recommendation and search scenarios. Compared to the best performance baseline, our full model achieves: **Recommendation Scenario:** +0.431 AUC improvement (0.69401 $\rightarrow$ 0.69832) with 5.89% loss reduction. **Search Scenario:** +0.422 AUC gain (0.72109 $\rightarrow$ 0.72531) accompanied by 6.94% loss decrease. Meanwhile, the incorporation of RSIA yields a measurable improvement in the AUC metric of the baseline model.

4.2.2 Ablation Study. To disentangle component contributions, we conduct surgical ablations on critical modules: There are three critical findings emerge: (1) **Dual MoE Contribution:** Removing the

Table 1: Offline performance comparison (Relative improvement over best baseline)

Model	AUC _{rec}	Loss _{rec}	AUC _{search}	Loss _{search}
ShareBottom	0.68457	0.00723	0.71745	0.00618
MMoE	0.69401	0.00679	0.72083	0.00598
PLE	0.69084	0.00693	0.71862	0.00634
STAR	0.69312	0.00681	0.71971	0.00605
PEPNet	0.69345	0.00684	0.72109	0.00576
ShareBottom+RSIA	0.69602	0.00686	0.72195	0.00598
MMoE+RSIA	0.69645	0.00688	0.72363	0.00568
PLE+RSIA	0.69584	0.00693	0.72262	0.00589
STAR+RSIA	0.69712	0.00666	0.72271	0.00570
PEPNet+RSIA	0.69713	0.00651	0.72409	0.00542
SuperRS (Ours)	0.69832	0.00639	0.72531	0.00536

Table 2: Component ablation analysis on both scenarios

Variants	AUC _{rec}	Δ	AUC _{search}	Δ
Full Model	0.69832	-	0.72531	-
w/o Dual MoE	0.69401	-0.431	0.71984	-0.547
w/o RSIA	0.69517	-0.315	0.72109	-0.422
w/o DFI	0.69688	-0.144	0.72315	-0.216

dual expert structure causes the most severe degradation, validating our core hypothesis about scenario-disparity modeling (2) **RSIA Necessity:** Disabling reciprocal attention reduces search-to-rec transfer efficiency, disproportionately impacting long-tail queries. (3) **DFI Gate Synergy:** The unified gating mechanism accounts proving its role in dynamic scenario adaptation.

4.2.3 Online A/B Test. We deploy SuperRS on the 1688 app serving 12 million daily active users. Our framework demonstrates: **+0.28% CTR** and **+0.15% CVR** uplift and **+3.14% GMV** increase. Meanwhile, a **+3.48% U-GMV** increase has been achieved through deepened insights into user needs.

5 Conclusion

This paper proposes SuperRS, a unified framework for integrating e-commerce search and recommendation systems through three core innovations: (1) a Dual-layer Multi-MoE (DualMoE) architecture that models scenario-specific patterns and cross-task interactions via hierarchical feature abstraction, and (2) a Reciprocal Scenario Interest Attention (RSIA) mechanism that enables bidirectional intent alignment by integrating target item relevance, historical behavior biases, and cross-scenario dependencies. (3) The framework employs Dynamic Feature Integration (DFI) gates for scenario-aware fusion and multi-objective optimization to balance expert specialization. The system achieves seamless synergy between search and recommendation functionalities. Both offline and online Experimental validation demonstrates statistically significant improvements.

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